Term Project – Final Report

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***Abstract*—This paper talks about the result for my term project**

Keywords—CRUM, enzyme, generalization, AI, cognitive

I. Abstract

Understanding how learning systems generalize from limited experience is a central problem in cognitive science. Generalization, as the ability to apply learned knowledge to novel situations, is closely tied to representation and inductive bias. This project investigates how the structure of training experience influences generalization in a computational system, using enzyme sequence learning as a testbed.

I constructed a proof-of-concept model that classifies methyltransferase sequences using k-mer–based representations. The model was trained under two conditions: a narrow experience condition, with limited exposure to enzyme types, and a diverse experience condition, with broader exposure. These conditions simulate different learning environments analogous to those studied in human cognition.

Results show that while both conditions achieve similar accuracy, the narrow condition is restricted to familiar categories, whereas the diverse condition generalizes across a broader range of enzyme types. These findings highlight a key cognitive principle: the structure of experience shapes the flexibility and transferability of learned representations. This work demonstrates how computational models in biological domains can serve as controlled environments for studying generalization and learning in cognitive systems.

II. Introduction

A. Importance

Understanding how systems generalize from limited experience to novel situations is a central problem in both cognitive science and artificial intelligence. In cognitive science, generalization underlies key abilities such as concept formation, categorization, and transfer of knowledge across contexts. In artificial intelligence, failures in generalization often lead to overfitting, where models perform well on familiar data but poorly on new inputs.

This question is particularly important in domains with extremely large and complex input spaces, such as protein sequence design. The space of possible protein sequences is vast, yet both biological systems and machine learning models are able to extract meaningful patterns from relatively limited data. Improving generalization in this domain has significant practical implications, including enzyme engineering, drug development, and biotechnology applications.

Methyltransferases, the focus of this study, are a class of enzymes that catalyze the transfer of methyl groups to target molecules. These modifications play critical roles in regulating biological processes, including gene expression, metabolism, and protein function. In pharmaceutical contexts, methyltransferases are particularly important because they can stabilize peptide-based therapeutics, helping to prevent degradation in the human body and improving drug efficacy. As a result, the ability to design or predict methyltransferase activity has direct implications for developing more robust and effective therapies.

Recent advances in AI have demonstrated strong performance in protein sequence modeling, but it remains unclear how these systems generalize and what kinds of training experience support robust and transferable learning. Prior work has emphasized the importance of inductive bias and training data structure in shaping model behavior. Therefore, investigating how the structure and diversity of training experience influence generalization is both a practical and theoretical problem.

B. Relation with cognitive science

This project is grounded in core cognitive science concepts, including representation, learning from experience, inductive bias, and generalization. From a cognitive science perspective, the training dataset can be interpreted as a form of experience, while the learned feature representations correspond to internal mental representations that support reasoning and categorization.

The project is informed by frameworks such as the CRUM perspective, which views cognition as computation over structured representations. In this system, sequence features (k-mer representations) serve as simplified analogs of internal representations, and the classification model performs computations over these representations.

Additionally, the task of predicting enzyme identity can be understood as a form of category learning, a well-studied area in cognitive science. The comparison between narrow and diverse training conditions directly relates to research on how different types of experience influence abstraction and transfer. Exposure to more varied input is hypothesized to support more flexible and generalizable representations, while limited experience may lead to more specific, less transferable knowledge.

By treating protein sequence learning as a computational testbed, this project examines how fundamental cognitive principles, such as the interaction between experience, representation, and inductive bias, shape generalization behavior in a learning system.

III. Model/Tool Design

A. Tools, platforms, frameworks used

The system was implemented in Python using standard machine learning libraries, including scikit-learn for model training and evaluation. Data processing was performed using custom scripts for parsing sequence data and extracting features.

B. *Cognitive science principles mapping*

This system operationalizes key cognitive science concepts:

Experience: training dataset composition

Representation: k-mer feature encoding of sequences

Inductive bias: model structure and feature choice

Generalization: performance on unseen sequences

Categorization: classification of enzyme identity

By manipulating the diversity of training data, the system simulates different learning environments analogous to those studied in human cognition.

C. *Principles resource*

The cognitive science principles guiding this system are drawn from both course materials and foundational literature in cognitive science and artificial intelligence.

First, the project is informed by the CRUM framework, which views cognition as computation over internal representations. In this system, protein sequence features (k-mer representations) operationalize internal representations, while the classification model performs computations over these representations.

Second, the project builds on theories of generalization and inductive bias in both cognitive science and machine learning. Research in cognitive science has shown that the structure of experience plays a critical role in shaping how learners generalize to novel situations. This idea is reflected in the experimental design, where different training conditions (narrow vs. diverse) simulate different learning environments.

Third, the work is influenced by cognitive science perspectives on category learning and transfer, which emphasize how exposure to varied examples supports abstraction and flexible knowledge. The classification task in this model can be interpreted as a form of categorization, where the system learns to map inputs to conceptual categories (enzyme types).

Finally, these ideas are reinforced by course lectures and readings on learning, representation, and generalization, which emphasize that intelligent behavior emerges from the interaction between experience, representation, and inductive constraints.

D. *System implementation*

Input: Protein sequences represented as amino acid strings

Output: Predicted enzyme identity (classification label)

Evaluation: Accuracy and classification performance on unseen sequences

Two conditions were defined: narrow experience condition, where model trained on a subset of enzyme types; diverse experience condition, where model trained on all available enzyme types

To increase data availability, sequences were divided into overlapping fragments, simulating repeated exposure to structured input. Sequences were then transformed into k-mer frequency vectors, capturing local amino acid patterns. This serves as a simplified model of internal representation in cognitive systems.

A Random Forest classifier was used due to its robustness on small datasets and ability to capture nonlinear relationships. Models were trained separately for each condition and evaluated on held-out data.

IV. Results

A. *Learning from the tool*

Building and evaluating this system revealed how the structure of training experience influences generalization behavior in a learning model. By implementing two conditions (narrow and diverse training), I am able to observe how differences in experience affect the model's ability to categorize enzyme sequences.

The model trained under the diverse condition achieved an overall accuracy of approximately 0.86 on the full test set. This result indicates that the system was able to correctly classify enzyme sequences across a wide range of enzyme types. In contrast, the narrow condition achieved an accuracy (~0.83), but only on a restricted subset of enzyme types encountered during training.

B. *Relation to research questions*

The research question asks how the diversity and structure of training experience affect a model's ability to generalize. The results show that while narrow training can produce strong performance within familiar categories, it does not support generalization beyond them. In contrast, the diverse condition demonstrates the ability to operate across a broader range of enzyme types, directly addressing the question of generalization.

C. *Findings and supporting evidence*

The system's behavior provides clear evidence that the structure of training experience influences generalization (Fig. I).

In my results, I used four values to precisely reflect the results:

- Precision measures how accurate the model's positive predictions are. It is defined as the proportion of correctly predicted instances among all instances predicted as a given class. High precision means that when the model predicts a class, it is usually correct.
- Recall measures how well the model captures all instances of a given class. It is defined as the proportion of correctly predicted instances among all actual instances of that class. High recall means the model successfully identifies most of the true examples.
- F1-score is the harmonic mean of precision and recall, providing a balanced measure of both. It is particularly useful when there is an uneven distribution of classes or when both false positives and false negatives are important. F1-score is defined as:

$$F1 = 2 * (precision * recall) / (precision + recall)$$

- Support refers to the number of actual instances of each class in the test dataset. It indicates how many samples were used to evaluate the model's performance for each category.

These metrics together provide a more complete picture of model performance than accuracy alone, especially in multi-class classification settings.

First, the model trained under the diverse condition achieved an overall accuracy of 0.86 on a test set containing all enzyme families (Table I-II). This demonstrates that the model learned representations capable of distinguishing among a wide range of methyltransferase families. In contrast, the narrow condition achieved a lower accuracy of 0.83 (Table III-IV), and this performance was measured on a restricted subset of enzyme families included in training. Despite being evaluated on an easier task, the narrow model did not outperform the diverse model.

Second, analysis of the classification reports reveals consistent performance patterns. In the diverse condition, many enzyme families (e.g., Aln, CYP, Lxm, Olv, SC, bmb) achieved near-perfect precision and recall, indicating that the model successfully captured distinguishing sequence features for these categories (Table I-II). However, other families (e.g., Bam, Bpum, Anc) showed lower precision and recall, suggesting that these classes may have more overlapping or less distinctive sequence patterns.

Third, the narrow model exhibits more constrained behavior. While it performs well on familiar enzyme families (e.g., Age, Leg, Stsp), it shows reduced recall for others (e.g., Pho), indicating less robust representation even within its limited domain (Table III-IV). Additionally, the narrow model cannot make predictions for enzyme families outside its training set, reflecting a lack of generalization beyond its experience.

Finally, comparing the two conditions reveals a key pattern: diverse training improves both performance and generalization scope. The diverse model not only achieves higher accuracy but also operates across a broader set of categories, while the narrow model remains restricted. This demonstrates that the diversity of training experience enhances the flexibility and transferability of learned representations.

Together, these findings provide strong evidence that training experience shapes generalization behavior, supporting the central research question of this project.

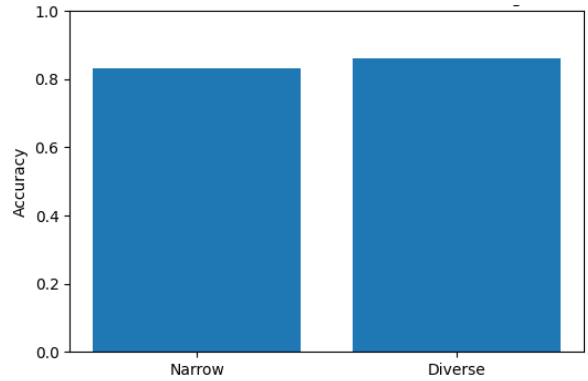

Figure I: Generalization: Narrow vs Diverse Training. It reflects comparison of classification accuracy between narrow and diverse training conditions. The diverse condition achieves higher accuracy despite operating on a broader set of enzyme families, indicating improved generalization from more diverse experience.

TABLE I. DIVERSE CLASSIFICATION REPORT

Enzyme Family	Value			
	Precision	***Recall***	***F1-score***	***Support***
Age	0.88	1.00	0.93	7
Aln	1.00	1.00	1.00	5
Anc	0.60	0.75	0.67	4
Bam	0.50	0.33	0.40	3
Bpum	0.50	0.33	0.40	3
CYP	1.00	1.00	1.00	3
Div	1.00	0.67	0.80	3
Lah	0.78	1.00	0.88	7
Leg	0.83	1.00	0.91	5
Lxm	1.00	1.00	1.00	3
Olv	1.00	1.00	1.00	6
PTT	0.67	0.50	0.57	4
Pho	1.00	0.80	0.89	5
SC	1.00	1.00	1.00	4
Stsp	0.83	1.00	0.91	5
Bmb	1.00	1.00	1.00	3
dpOp	0.88	0.75	0.82	9

TABLE II. DIVERSE CLASSIFICATION SUMMARY

Title	Value			
	Precision	***Recall***	***F1-score***	***Support***
Accuracy	-	-	0.86	79
Macro avg	0.85	0.83	0.83	79
Weighted avg	0.86	0.86	0.85	79

TABLE III. NARROW CLASSIFICATION REPORT

Enzyme Family	Value			
	Precision	***Recall***	***F1-score***	***Support***
Age	1.00	1.00	1.00	7
Anc	0.67	0.50	0.57	4
Leg	1.00	1.00	1.00	5
PTT	0.75	0.75	0.75	4
Pho	1.00	0.40	0.57	5
Stsp	0.83	1.00	0.91	5
Bmb	1.00	0.67	0.80	3
dpOp	0.69	1.00	0.82	9

TABLE IV. NARROW CLASSIFICATION SUMMARY

Title	Value			
	Precision	***Recall***	***F1-score***	***Support***
Accuracy	-	-	0.83	42
Macro avg	0.87	0.79	0.80	42
Weighted avg	0.86	0.83	0.82	42

V. Discussion

A. Implications for cognitive science

This work demonstrates how biological sequence learning can serve as a computational model of category learning and generalization in cognitive systems.

The results provide strong evidence that the structure of experience plays a critical role in shaping generalization, a central question in cognitive science. The diverse training condition achieved higher accuracy (0.86) than the narrow condition (0.83) while also operating across a broader set of enzyme families. This indicates that exposure to more varied experience supports the formation of more flexible and transferable representations.

This finding aligns with theories of inductive bias and representation, which suggest that learning systems extract structure from experience based on both input diversity and internal constraints [1]. In cognitive science, similar effects are observed in category learning, where exposure to more varied examples enables learners to form more abstract concepts that generalize across contexts [2].

The comparison between narrow and diverse conditions also highlights a key cognitive distinction between performance and generalization. The narrow model performs reasonably well within its limited domain but fails to extend beyond it, reflecting a form of overfitting. In contrast, the diverse model demonstrates broader applicability, suggesting that its internal representations capture more generalizable structure. This parallels human cognition, where broader experience supports transfer and abstraction [3].

B. System importance

This system matters because it provides a controlled computational testbed for studying cognitive principles in a complex, high-dimensional domain [4]. Protein sequences are structurally rich and biologically meaningful, making them an ideal environment for examining how learning systems generalize from limited data.

Importantly, the system demonstrates that even relatively simple models can exhibit key cognitive phenomena:

- sensitivity to experience structure
- trade-offs between specificity and generalization
- dependence of learning outcomes on representation

By grounding these phenomena in a real-world biological domain, the project bridges cognitive science theory and applied machine learning, showing how abstract concepts like representation and inductive bias manifest in practical systems.

C. Real-world applications

The findings have implications for both artificial intelligence and biotechnology.

In AI, the results emphasize that training data diversity is critical for robust generalization, reinforcing the need to carefully design datasets and learning environments. This insight applies broadly to domains such as natural language processing, computer vision, and scientific modeling [5].

In biotechnology, improved generalization in enzyme sequence models could support the design and prediction of methyltransferase activity, which is important for stabilizing peptide-based therapeutics. By enabling models to generalize across enzyme families, such systems may help identify novel enzymes with desirable properties for drug development.

More broadly, this work suggests that improving learning systems requires not only better models, but also a deeper understanding of how experience shapes representation and generalization.

To summarize, these results suggest that generalization in both artificial and biological systems emerges from the interaction between experience structure and representation, reinforcing a central principle of cognitive science.

VI. Conclusion

A. Learning

This project investigated how the structure and diversity of training experience affect generalization in a computational learning system. By comparing narrow and diverse training conditions, I found that broader experience enables models to operate across a wider range of enzyme types, while limited experience restricts performance to familiar categories.

B. Key insights

The key insight from this work is that generalization is shaped by experience structure, not just model complexity or training size. From a cognitive science perspective, this reinforces the importance of representation and inductive bias in learning.

Future work could expand this approach by incorporating larger datasets, more sophisticated models, and deeper analysis of learned representations. Additionally, extending this framework to other biological or cognitive domains could further illuminate how learning systems generalize from experience.

VII. Limitations

A. *Limitations in current system*

This system has several limitations that should be considered.

First, the dataset is relatively small, which limits the robustness and generalizability of the results. Although sequence fragmentation increases the number of training samples, the underlying diversity of the dataset remains constrained.

Second, the feature representation is simplified. The use of k-mer frequency vectors captures local sequence patterns but does not account for higher-order structural or biochemical properties of proteins, which may be important for function.

Third, the comparison between narrow and diverse conditions is not perfectly symmetrical. The narrow condition is evaluated on a restricted subset of classes, making direct quantitative comparison with the diverse condition challenging.

Fourth, the model itself is relatively simple and does not represent state-of-the-art approaches in protein modeling. The goal of this system was interpretability and conceptual clarity rather than performance optimization.

B. *Future improvements*

Future work could address these limitations by:

- Expanding the dataset to include more enzyme sequences and families
- Using more expressive representations (e.g., embeddings from protein language models)
- Implementing more advanced models to capture complex sequence relationships
- Designing more controlled experimental setups to better isolate the effects of experience structure

These improvements would allow for a more rigorous investigation of generalization in both cognitive and biological contexts.

VIII. Code submission

Link to Github:

https://github.com/hliang305/cs6795-enzyme-generalization

References

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